EFA Annual Meeting 2026 --- Poster Session

Hedge Fund Trading and Sovereign Bond Yield Sensitivity

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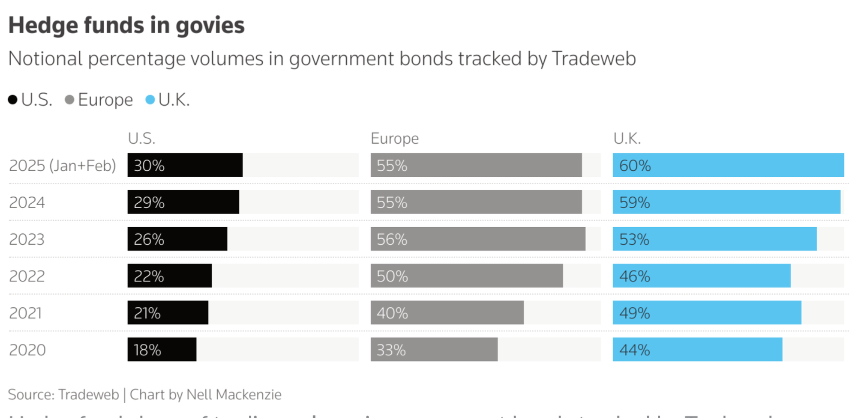
Hedge funds hold a growing footprint in government bond markets



Hedge fund share of trading volume in government bonds tracked by Tradeweb.

- Their share of trading volume keeps rising since 2020: in Europe from a third to **more than half** of volumes.
- They finance sovereign positions in the **repo market**, so leverage ties their funding to the value of the collateral.
- Granular evidence on how this footprint affects price formation outside of crises is scarce.

Decrease sensitivity

Arbitrage corrects mispricings.
Prices are pushed *toward* fundamentals.
Gromb and Vayanos (2002), Cao et al. (2018)

Increase sensitivity

Leverage and funding fragility force trades.
Prices are pushed *away* from fundamentals.
Shleifer and Vishny (1997), Brunnermeier and Pedersen (2009)

► **Research question: does the leveraged positioning of hedge funds amplify sovereign bond yield sensitivity to shocks — and under what conditions?**

Data and empirical design

Hedge fund repo positions

ECB SFTDS

Transaction-level euro area repo data. German and Italian sovereigns, Jan 2021 – Nov 2025. 152 identified offshore hedge funds. Repo $=$ long, reverse repo $=$ short.

Bond characteristics

Bloomberg

Daily yields, duration, bid-ask spread, amount issued, and cheapest-to-deliver status for each individual bond.

Exogenous shocks

EA-MPD (Altavilla et al. 2019)

Intraday 2-year OIS change in the ECB event window. The same shock hits every bond: identification comes from cross-sectional differences in positioning.

$$\Delta y_{i,t} \;=\; \beta_1\,(\mathrm{HF}_{i,t-1} \times \mathrm{Shock}_t) \;+\; \beta_2\,\mathrm{HF}_{i,t-1} \;+\; \beta_3\,\mathrm{Shock}_t \;+\; \gamma\,\mathbf{X}_{i,t-1} \;+\; \theta_k \;+\; \varepsilon_{i,t}$$

β_1 is the **additional** yield sensitivity to the shock when hedge funds are positioned in a bond, measured by a presence dummy or by intensity (five-day net position / amount outstanding).

Identification. Hedge funds select liquid, large, longer-duration and cheapest-to-deliver bonds. Therefore: duration $\times$ country $\times$ date fixed effects compare similar bonds of the same issuer on the same day; a matched design pairs each hedge fund bond with its closest twin; a dealer-level panel shows the trigger is not dealer behavior.

152	477,001	38	53%	3%
hedge funds	bond-days	ECB announcements	of bond-days with HF activity	of an issue held when active

How leveraged positioning amplifies yields

Starting point (Adrian and Shin, 2014): leveraged intermediaries manage their balance sheet against the risk environment — the same rule produces opposite position adjustments depending on the sign of the shock.

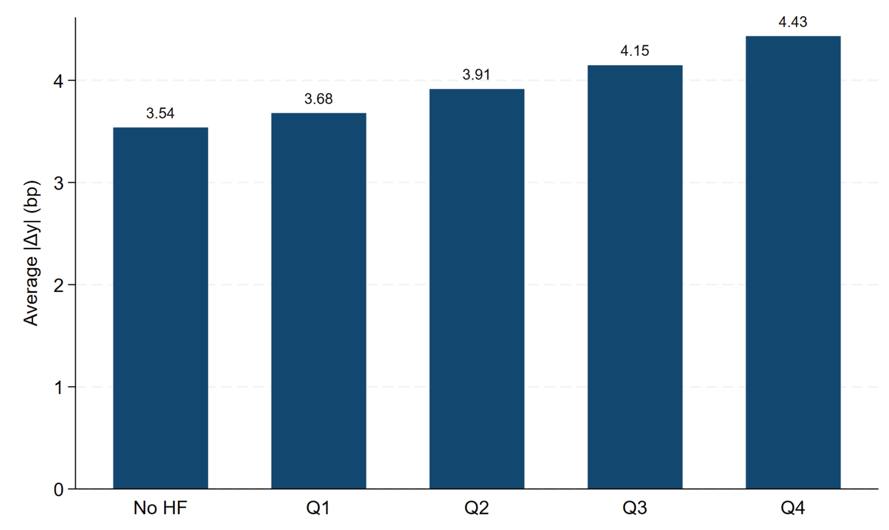
Constraining ← **Shock meets a pre-determined leveraged position** → **Relaxing**

- **Constraining**: price moves *against* the position — capital buffer erodes → fund **reduces** exposure
- **Relaxing**: price moves *in favor* — constraint loosens → fund **expands** exposure

A numerical example (effective margin 2%, duration 7y, sector holds 3% of the issue): 6 bp surprise $\to$ price -0.42% $\to$ capital buffer -21% $\to$ trading pressure 0.63% of the issue $\to$ $+3.15$ **bp** extra yield move.

► **In both states the resulting trading pushes yields further in the direction of the initial shock.**

Finding 1: Hedge funds increase the volatility of bond yields on shock days



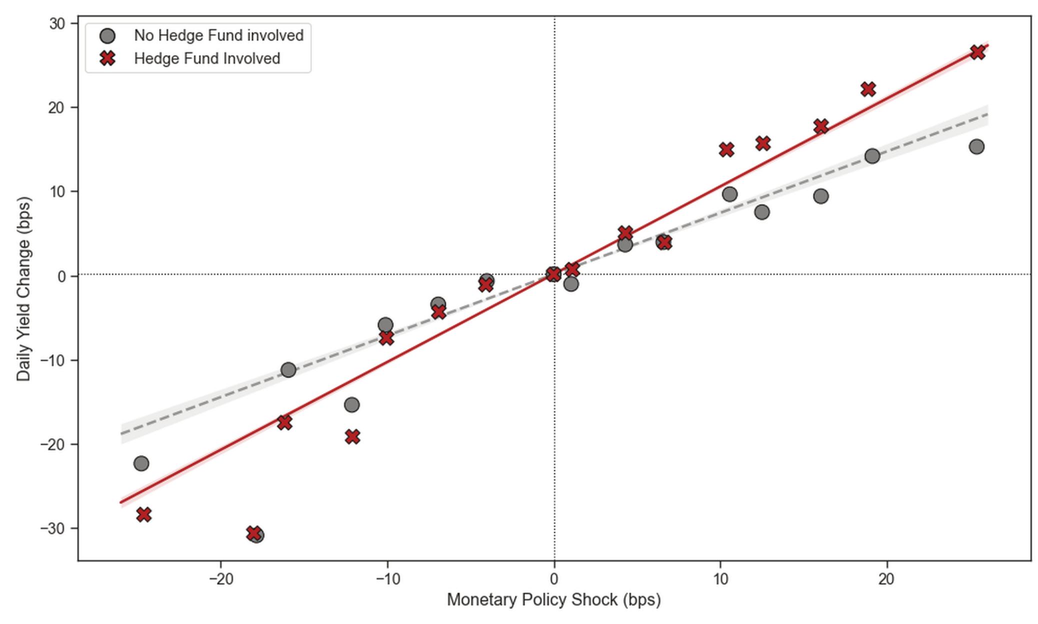
Average $|\Delta\text{yield}|$ (bp) on non-shock days by quartile of hedge fund intensity, next to bonds without hedge funds.

Hedge funds do select structurally more volatile bonds. But among **comparable bonds** (duration $\times$ country $\times$ date cells) the level differential disappears, while the interaction with the shock survives:

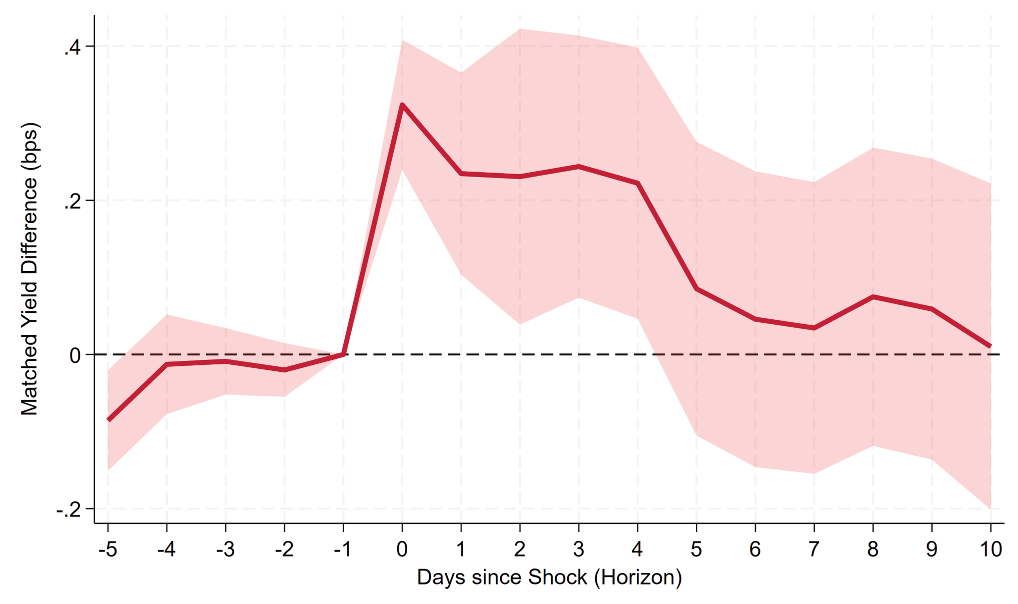
$\lvert\Delta$yield$\rvert$ (bp)	Raw	Comparable bonds
HF involved $\times$ $\lvert$Shock$\rvert$	**0.240*****	**0.246*****
HF involved (level)	0.459***	-0.098

► **The excess volatility is shock-conditional — what changes is how yields react to shocks.**

Finding 2: Hedge fund trading raises the sensitivity of yields to shocks



Binned scatter of daily yield changes against monetary policy surprises, with and without active hedge fund positions.



Local projections, matched sample (Jordà 2005): flat before the shock, divergence on impact, significant for $\approx$ 4 days, then reversion.

Δ yield (bp)	Baseline	Matched	Intensity
HF involved $\times$ Shock	**0.291*****	**0.302*****	
HF intensity $\times$ Shock			**0.164*****
HF (level)	-0.022	-0.002	-0.011

Duration $\times$ country $\times$ date FE resp. matched-pair FE. Within the same bond, a larger position produces a larger reaction — pure selection cannot explain this.

$>$ 25%

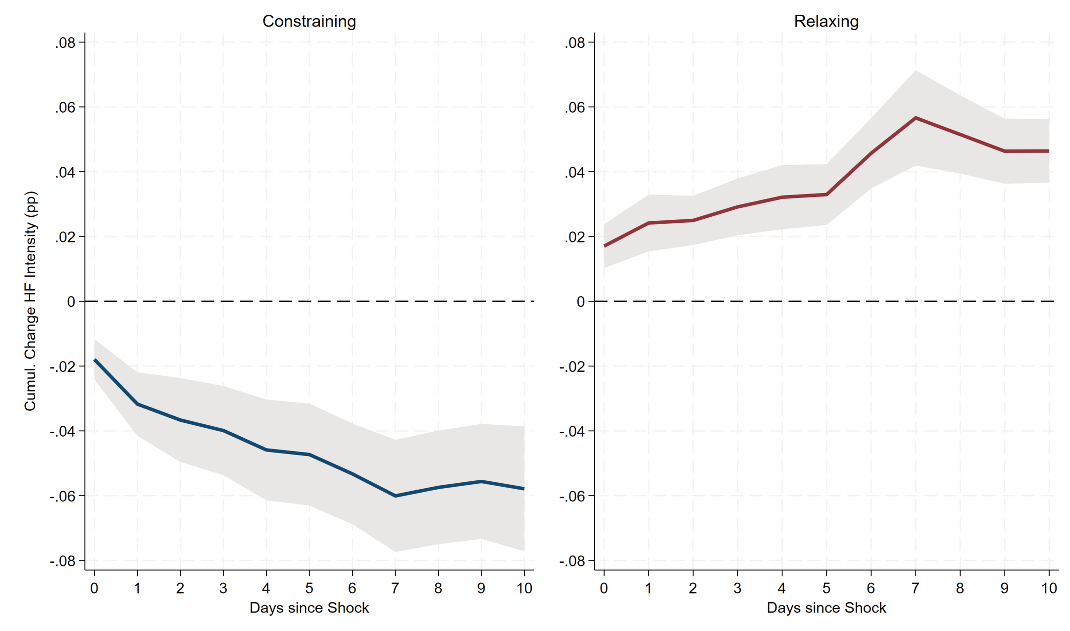
of the shock-day yield movement in hedge-fund-traded bonds reflects **positioning**, not the news itself.

► **Temporary overshooting that other capital gradually absorbs — amplification, not price discovery.**

Finding 3: The amplification holds whether shocks help or hurt the position

Δ yield (bp)	Constraining	Relaxing
HF intensity $\times$ Shock	**0.140*****	**0.224*****
HF intensity (level)	-0.016	-0.009

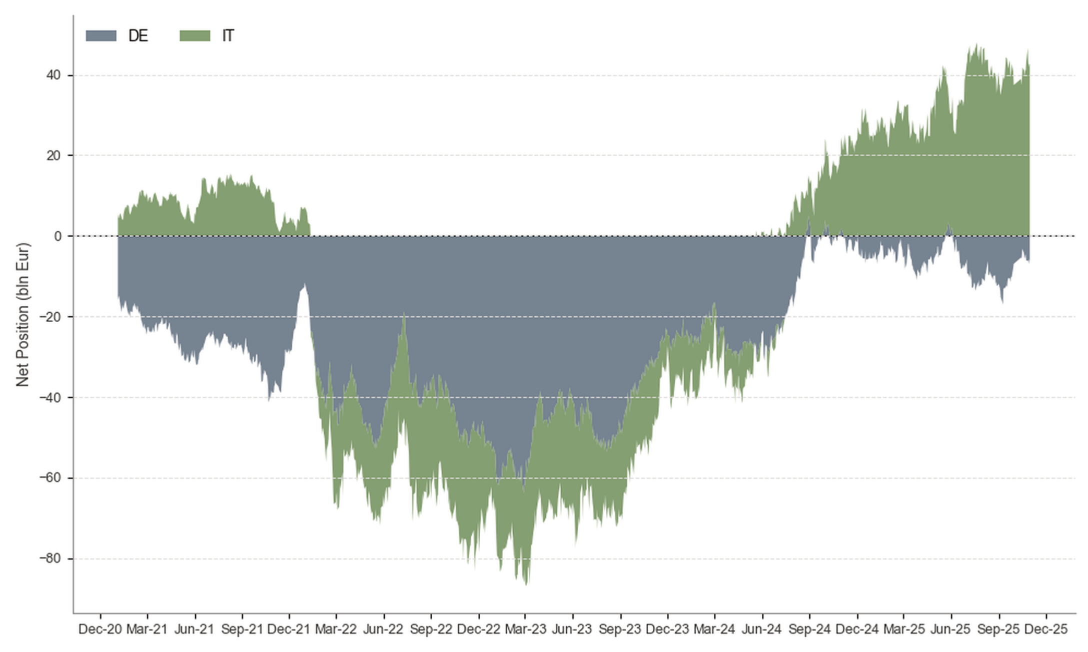
A Wald test cannot distinguish the two regimes ($p = 0.16$): the amplification is **symmetric**.



Overnight repo positions after the shock: funds contract in the constraining regime and expand in the relaxing regime, as the mechanism predicts.

► **Amplification is not only a crisis phenomenon — it operates whenever shocks meet positions.**

Finding 4: The directionality of the book governs the strength



Aggregate net repo positions in German (DE) and Italian (IT) sovereigns. Around the hiking cycle: long IT, short DE (hedged book). During it: one-sided short (directional book).

A fund manages one pool of capital against its whole book: a duration-hedged book barely loses capital when rates move, a directional book absorbs the full hit.

Δ yield (bp), rel. to no HF	Coefficient
Hedged book $\times$ Shock	0.098***
Directional book $\times$ Shock	**0.363*****
Q1 most hedged $\times$ Shock	0.057
Q2 $\times$ Shock	0.137***
Q3 $\times$ Shock	0.361***
Q4 most directional $\times$ Shock	**0.367*****

► **The amplification rises monotonically with directionality and vanishes when the book is hedged.**

Conclusion: what this means for risk monitoring

- Hedge fund positioning amplifies the yield response to shocks: **more than a quarter** of the shock-day move, reverting within days.
- The amplification is **symmetric** across constraining and relaxing states — consistent with constant-leverage targeting.
- Size and presence alone are not sufficient risk metrics, and amplification is **not confined to stress episodes**.
- The **directionality** of the book is a key risk metric beyond position size.

Robust to: dealer $\times$ date fixed effects (0.248–0.294***, no scaling with dealer CDS stress) · GARCH-standardized yields · sovereign credit (CDS) shocks (0.225***) · French and Spanish bonds · placebo with shifted shock dates (-0.030, n.s.) · leave-one-out across announcements · randomization inference ($p < 0.001$).

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References: Adrian–Shin (2014) · Altavilla et al. (2019) · Jordà (2005) · Jarocinski–Karadi (2020) · Barth–Kahn (2025) · Kruttli et al. (2025) · Gromb–Vayanos (2002) · Brunnermeier–Pedersen (2009)